

$$= \int_0^1 A x^2 (1-x)^2 dx;$$

for value of A;

we know;

$$\int_0^1 A x^2 (1-x)^2 dx = 1$$

$$\Rightarrow A \int_0^1 x^2 (1 - 2x + x^2) dx = 1$$

$$\Rightarrow A \int_0^1 (x^2 - 2x^3 + x^4) dx = 1$$

$$\Rightarrow A \left[\frac{x^3}{3} - \frac{2x^4}{4} + \frac{x^5}{5} \right]_0^1 = 1$$

$$\Rightarrow A = \frac{1}{\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right)}$$

$$= \frac{10}{30}$$

$$\Rightarrow A \int_0^1 x^2 (1-x)^2 dx = 1$$

$$\Rightarrow A \int_0^1 x^{3-1} (1-x)^{2-1} dx = 1$$

$$\Rightarrow A \frac{\Gamma_3 \Gamma_3}{\Gamma_{3+3}} = 1$$

$$\Rightarrow A \frac{2! 2!}{5!} = 1$$

$$A = 12$$

Now.

$$E(x) = \int_0^1 \cancel{12} x \cdot f(x) dx$$

$$= \int_0^1 12 \cdot x^2 (1-x)^2 dx$$

$$= 12 \int_0^1 x^{3-1} (1-x)^{3-1} dx$$

$$= 12 \cdot \frac{\Gamma(3) \cdot \Gamma(3)}{\Gamma(6)} \{\beta(3,3)\}$$

$$= \cancel{12} \cdot \frac{2 \cdot 2}{5 \times \cancel{24}}$$

$$= \frac{2}{5}$$

$$E(x^2) = \int_0^1 x^2 \cdot f(x) dx$$

$$= \int_0^1 12 x^3 (1-x)^2 dx$$

$$= 12 \int_0^1 x^{4-1} (1-x)^{3-1} dx$$

$$= 12 \beta(4,3)$$

$$= 12 \cdot \frac{\Gamma(4) \Gamma(3)}{\Gamma(4+3)} = 12 \cdot \frac{6 \times 2}{\cancel{6} \times 5 + \cancel{24}}$$

$$= \frac{1}{25}$$

Ans:

$$V(x) = E(x^2) - [E(x)]^2$$

$$= \frac{1}{25} - \left(\frac{2}{5}\right)^2$$

$$= \frac{1}{25} - \frac{4}{25} = 0$$

$$\frac{1}{25} - \frac{4}{25} = 0$$

Imp. Ex:

A coin is tossed until a head appears. What is the expectation of the number of tosses required to get first head?

Soln:

Let, $x = \text{no. of tosses in a coin } \{H, T\}$

Prob. of Head: $P(H) = \frac{1}{2}$; Prob. of Tail: $P(T) = 1 - \frac{1}{2} = \frac{1}{2}$

x	1	2	3	...
$P(x)$	$\frac{1}{2}$	$\left(\frac{1}{2}\right)^2$	$\left(\frac{1}{2}\right)^3$	...

$$E(x) = \sum_{x=1}^{\infty} x \cdot P(x=x) = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} + \dots$$

$$= \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \dots$$

$$\therefore \text{gen. form} = \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{1}{2} + \dots$$

$$= 1 + 3 \left(\frac{1}{2}\right)^3$$

$$= 1 \cdot \frac{1}{2} + 2 \cdot \left(\frac{1}{2}\right)^2 + 3 \cdot \left(\frac{1}{2}\right)^3 + \dots$$

5. $\frac{1}{2} = x$ then; generating fun will be:

$$= 1 \cdot x + 2 \cdot x^2 + 3 \cdot x^3 + \dots$$

$$= x + 2x^2 + 3x^3 + \dots$$

$$= x (1 + 2x + 3x^2 + \dots)$$

$$= x \left\{ \frac{1}{(1-x)^2} \right\} \left[\because \frac{1}{1-x^2} = 1 + x + x^2 + \dots \right]$$

$$|x| < 1$$

$$= x \cdot (1-x)^{-2}$$

where, $x = \frac{1}{2}$

$$\therefore E(x) = \frac{1}{2} \left(1 - \frac{1}{2}\right)^{-2}$$

$$= \frac{1}{2} \left(\frac{1}{2}\right)^{-2}$$

$$= \frac{1}{2} \cdot \frac{1}{\left(\frac{1}{4}\right)}$$

$$= \frac{4}{2}$$

$$= 2$$

$$\text{or; } E(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$$

$$\frac{E(x)}{2} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots$$

$$E(x) - \frac{E(x)}{2} = \frac{1}{2} +$$

$$\left(\frac{2}{4} - \frac{2}{4}\right) + \left(\frac{3}{8} - \frac{2}{8}\right)$$

$$+ \left(\frac{4}{16} - \frac{3}{16}\right) + \dots$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$+ \dots$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{2}} \quad (\text{by infinite sum of G.S.})$$

$$\frac{2}{1-x}, x < 1, \quad \frac{2}{x-1}, x > 1$$

→ Joint Probability Distribution = list of

→ The probability distribution of two-dimensional R.V.s (X, Y) is called Bivariate Distribution.

eg: (i) Distribution of Height and weight of a group of students.

(ii) Distribution of Income and Expenditure of a group of individuals.

Note: For Bivariate Distribution,

Both R.V.s must be of same type.

i.e., both are either discrete or continuous R.V.s

→ Joint Probability Distribution of Discrete R.V.s X and Y :

→ Joint probability mass function of X and Y .

Let, X and Y are two discrete R.V.s with joint probability function $P(X, Y)$ or $P(X=x, Y=y)$ $\forall x, y$.

Then, the joint prob. function $P(X=x, Y=y)$ is called joint probability mass function that satisfies the following condition:

$$i) P(X=x, Y=y) \geq 0 \rightarrow \text{Non-negative}$$

$$ii) \sum_x \sum_y P(X=x, Y=y) = 1 \rightarrow \text{Total prob.} = 1$$

eg; let X and Y are two discrete r.v.s with joint prob. mass function.

$X \backslash Y$	1	2	3
1	0.1	0.2	0.15
2	0.2	0.3	0.1
3	0.1	0.15	0.2

$$\begin{aligned}
 P(X=2, Y=1) &= 0.1 \\
 P(X=2, Y=2) &= 0.2 \\
 P(X=2, Y=3) &= 0.1 \\
 P(X=3, Y=1) &= 0.15 \\
 P(X=3, Y=2) &= 0.3 \\
 P(X=3, Y=3) &= 0.15
 \end{aligned}$$

Joint
prob.

as a whole
joint prob.
distribution

* Marginal Distribution
(Marginal Prob. Distr.)

Let, X and Y are two discrete r.v.s with joint pmf $P(X=x, Y=y)$

Then, the marginal Probability Distribution of X is denoted by $P(X=x)$ and of Y is denoted by $P(Y=y)$

$$P(X=x) = \sum_y P(X=x, Y=y)$$

Similarly, the marginal prob. distribution of Y is denoted by $P(Y=y)$ and is calculated as;

$$P(Y=y) = \sum_x P(X=x, Y=y)$$

eg,

$Y \backslash X$	2	3
1	0.1	0.15
2	0.2	0.3
3	0.1	0.15

→ Find the marginal distribution of X & Y .

We know:

$$P(X=x) = \sum_{y=1}^3 P(X=x, Y=y)$$

Marginal distribution of X :

for $x=2$;

$$P(X=2) = \sum_{y=1}^3 P(X=2, Y=y)$$

$$= P(X=2, Y=1) + P(X=2, Y=2) +$$

$$P(X=2, Y=3) \text{ (or } P(2,1) + P(2,2) + P(2,3))$$

$$= 0.1 + 0.2 + 0.1$$

$$= 0.4$$

for $x=3$;

$$P(X=3) = \sum_{y=1}^3 P(X=3, Y=y)$$

use write in
ordered pair
of (x,y)

$$= P(3,1) + P(3,2) + P(3,3)$$

$$= 0.15 + 0.3 + 0.15$$

$$= 0.6.$$

Similarly, marginal distribution of Y .

$$P(Y=y) = \sum_{x=2}^3 P(X=x, Y=y)$$

for $Y=1$;

$$P(Y=1) = P(2,1) + P(3,1)$$

$$= 0.1 + 0.15$$

$$= 0.25.$$

for $Y=2$;

$$P(Y=2) = P(2,2) + P(3,2) = 0.2 + 0.3 = 0.5$$

for $Y=3$;

$$P(Y=3) = P(2,3) + P(3,3) = 0.1 + 0.15 = 0.25$$

Σ, marginal distribution of X

$X=x$	$P(X=x)$
2	0.4
3	0.6

$Y=y$	$P(X=Y)$
1	0.25
2	0.5
3	0.25

Conditional Probability Distribution:

Let X and Y are two discrete R.V.s with joint pmf $P(X=x, Y=y)$ or $P(x, y)$; $\forall x, y$.

Also, let $P(X=x)$ and $P(Y=y)$ are marginal distribution of X and Y respectively. Then, the conditional probability distribution of $X=x$ given that $Y=y$ is denoted by $P(X=x/Y=y)$

and is calculated as;

$$P(X=x/Y=y) = \frac{P(X=x \cap Y=y)}{P(Y=y)}$$
$$= \frac{P(X=x, Y=y) \rightarrow \text{joint}}{P(Y=y) \rightarrow \text{marginal}}$$

provided that $P(Y=y) \neq 0$

Similarly, the conditional probability distribution of $Y=y$ given $X=x$ which is denoted by $P(Y=y/X=x)$ & is calculated as;

$$P(Y=y/X=x) = \frac{P(X=x, Y=y)}{P(X=x)}; P(X=x) \neq 0$$

Q.9

$Y \backslash X$	2	3
1	0.1	0.15
2	0.2	0.5
3	0.1	0.15

(i) Find the marginal distribution of X & Y

(ii) Find the conditional prob. of $X=x/Y=y$

ii) Find the conditional prob- of $P(Y=2/X=2)$.

Conditional prob- of $X=x$ given $Y=1$:

$$P(X=x/Y=1) = \frac{P(X=x, Y=1)}{P(Y=1)}$$

for $x=2$;

$$\begin{aligned} P(X=2/Y=1) &= \frac{P(X=2, Y=1)}{P(Y=1)} \\ &= \frac{0.1}{0.25} = 0.4 \end{aligned}$$

for $x=3$;

$$\begin{aligned} P(X=3/Y=1) &= \frac{P(X=3, Y=1)}{P(Y=1)} \\ &= \frac{0.15}{0.25} \\ &= 0.6 \end{aligned}$$

Similar. Conditional prob- of $Y=y$ given $X=2$:

$$P(Y=y/X=2) = \frac{P(X=2, Y=y)}{P(X=2)}$$

for $y=1$;

$$P(Y=1/X=2) = \frac{P(X=2, Y=1)}{P(X=2)} =$$

for $y=2$;

$$P(Y=2/X=2) = \frac{P(X=2, Y=2)}{P(X=2)} = \dots$$

for $y=3$;

$$P(Y=3/X=2) = \frac{P(X=2, Y=3)}{P(X=2)} = \dots$$

Condition for Independence:

Two discrete r.v.s X and Y are called independent; if

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$$

Otherwise, not independent.

Alternatively,

Two r.v.s X and Y are called independent event if;

$$P(X=x/Y=y) = P(X=x).$$

$$\text{Also, } P(Y=y/X=x) = P(Y=y)$$

Otherwise not independent.

9) $\rightarrow$

$X \backslash Y$	2	3
1	0.1	0.15
2	0.2	0.5
3	0.1	0.15

① Are X and Y independent?

Soln

$$P(X=2, Y=1) = 0.1$$

$$P(X=2) = 0.4 \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{ we calculated it before.}$$

$$P(Y=1) = 0.25$$

Now,

$$P(X=2, Y=1) = 0.4 \times 0.25 \\ = 0.1$$

$$\therefore P(X=2, Y=1) = P(X \neq 2) \cdot P(X=1) \quad \therefore X \text{ \& } Y$$

are independent.

Alternatively:

$$P(X=2 | Y=1) = 0.4$$

$$P(X=2) = 0.4$$

$$\therefore P(X=2 | Y=1) = P(X=2) \quad \text{So, } X \text{ \& } Y \\ \text{are independent.}$$

Joint Distribution Function of Discrete X

Let X and Y are two discrete r.v.s with joint pmf $P(X=x, Y=y)$;
 $\forall x, y$.

Then the joint distribution function of X and Y is denoted by $F(x, y)$ and is calculated as;

$$F(x, y) = P(X \leq x, Y \leq y)$$

$$= \sum_{-\infty}^x \sum_{-\infty}^y P(X=x, Y=y)$$

Q)

Y \ X	(1)	3
(1)	0.1	0.15
(2)	0.2	0.3
(3)	0.1	0.15

① Find $F(2, 3)$ and $F(3, 2)$.

$$F(2, 3) = P(X \leq 2, Y \leq 3)$$

$$= 0.1 + 0.2 + 0.1 \quad \text{2 intersected values}$$

$$F(3, 2) = P(X \leq 3, Y \leq 2)$$

$$= 0.2 + 0.15 + 0.2 + 0.3 =$$

Y \ X	(2)	(3)
(1)	0.1	0.15
(2)	0.2	0.3
3	0.1	0.15

Q) The joint prob. distribution of two discrete r.v.s X and Y is given as:

$Y \backslash X$	1	2	3
1	$4/32$	$3/32$	$2/32$
2	$7/32$	$5/32$	$1/32$
3	$2/32$	$2/32$	$6/32$

i) Find the Marginal distribution of X and Y .

ii) Find the conditional distribution of $X=x$ given $Y=2$ and $Y=y$ given $X=3$.

iii) Are X and Y independent?

iv) Find $F(2,3)$, $F(2,2)$ and $F(3,1)$.

v) Find $E(X)$, $E(Y)$, $V(X)$ and $V(Y)$.

vi) Find covariance of X and Y , i.e., $\text{Cov}(X, Y)$

⑦ $\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$

Here, $E(XY) = \sum_x \sum_y x \cdot y \cdot P(X=x, Y=y)$

$$= \underset{X}{1} \times \underset{Y}{1} \times \underset{\substack{\text{joint prob.}}}{4/32} + \underset{X}{1} \times \underset{Y}{2} \times \underset{P(X,Y)}{7/32} + \underset{X}{2} \times \underset{Y}{3} \times \underset{P(X,Y)}{2/32}$$

$$+ 2 \times 1 \times 3/32 + 2 \times 2 \times 5/32 + 2 \times 3 \times 2/32 +$$

$$3 \times 1 \times 2/32 + 3 \times 2 \times \frac{1}{32} + 3 \times 3 \times 6/32$$

Joint Probability Distribution of Continuous R.Vs X and Y

→ Joint probability density function of X and Y:

Let, X and Y are two continuous R.Vs with joint probability function $f(x, y)$.

Then, the joint probability function $f(x, y)$ is called joint probability density function which satisfies the following conditions:

i) $f(x, y) \geq 0$; i.e., Non-negative.

ii) $\int \int f(x, y) dy dx = 1 \Rightarrow$ Total prob = 1.

$$\Rightarrow \int \left[\int f(x, y) dy \right] dx = 1 \text{ or}$$

eg. Let X and Y are two continuous R.Vs with joint ~~pdf~~ pdf; $f(x, y) = 4xy$; $0 < x < 1, 0 < y < 1$ or function

Verify the given joint function is a joint pdf.

Soln $f(x, y) \geq 0$ which is obvious as x & y range is 0 to 1.

To verify the given joint function $f(x, y)$ is joint pdf. we need to show that the total prob. = 1.

i.e.,
$$\int_x \int_y f(x, y) dy dx = 1$$

$$= \int_0^1 \int_0^1 4xy dy dx$$

$$= \int_0^1 \left[\frac{4xy^2}{2} \right]_0^1 dx$$

$$= \int_0^1 2x dx$$

$$= 1 \#$$

$\therefore$ Total prob. = 1. Hence, verified # that given joint function is a joint pdf.

Marginal Density Function

Let, X and Y are two continuous R.Vs with joint pdf $f(x, y)$.

Then, the marginal density function of X is denoted by $f(x)$ and is calculated as:

$$f(x) = \int_y f(x, y) dy$$

Similarly, marginal density function of Y :

$$f(y) = \int_x f(x, y) dx$$

eg, ~~if~~ joint pdf. $f(x, y) = 4xy$; $0 < x < 1$, $0 < y < 1$

Find $f(x)$ & $f(y)$. i.e., marginal density function of x .

Solⁿ

We know,

$$f(x) = \int_y f(x, y) dy$$

$$= \int_0^1 4xy dy$$

$$= 4x \left[\frac{y^2}{2} \right]_0^1$$

$$= 2x \quad ; \quad 0 < x < 1 \text{ (don't forget}$$

to write range)

&

$$f(y) = \int_x f(x, y) dx$$

$$= \int_0^1 4xy dx$$

$$= 4y \int_0^1 x dx$$

$$= 4y \left[\frac{x^2}{2} \right]_0^1$$

$$= 2y \quad ; \quad 0 < y < 1$$

~~Marginal Den~~

Iron Maiden → Fear of the Dark.

Conditional density function

Let X and Y are two Continuous R.V.s with joint pdf. $f(x, y)$.

Also, let $f(x)$ & $f(y)$ be the marginal density functions of x & y respectively. Then,

conditional density function of x given the y , denoted by $f(x|y)$, or $f_{x|y}(x|y)$ &

~~$f(x|y)$~~ is calculated as;

$$f(x|y) = \frac{f(x, y)}{f(y)} ; \text{ provided that } f(y) \neq 0$$

Similarly, the conditional density function of y given that x , denoted by $f(y|x)$ or $f_{y|x}(y|x)$ is calculated as;

$$f(y|x) = \frac{f(x, y)}{f(x)} ; \text{ provided that } f(x) \neq 0$$

e.g., let x and y be two R.V.s and the joint pdf is given as;

$$f(x, y) = 4xy ; 0 < x < 1, 0 < y < 1$$

Find the conditional density function of x given y . Also calculate conditional density function of y given x .

Soln. we know $f(x|y)$ i.e., conditional density function of x given y .

$$\Rightarrow f(x/y) = \frac{f(x, y)}{f(y)}$$

$$= \frac{4xy}{2y}$$

$$= 2x ; 0 < x < 1$$

Similarly,

$\hookrightarrow$ as x & y are independent

$$f(y/x) = \frac{f(x, y)}{f(x)}$$

$$= \frac{4xy}{2x}$$

$$= 2y ; 0 < y < 1$$

Conditions for independence

Two continuous R.V.s X and Y are independent if,

$$f(x, y) = f(x) \cdot f(y)$$

otherwise not independent.

Alternatively,

$$f(x/y) = f(x)$$

$$\text{or, } f(y/x) = f(y)$$

otherwise not independent

Joint Distribution Function of Continuous R.V.s X and Y

Let X and Y are two continuous R.V.s with joint prob. density $f(x, y)$.

Then, the joint distribution function of X and Y is denoted by $F(x, y)$ and is calculated as;

$$F(x, y) = P(X \leq x, Y \leq y)$$

$$= \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx$$

eg; Let X and Y be two R.V.s with joint pdf given as;

$$f(x, y) = 4xy; 0 < x < 1, 0 < y < 1$$

Now, calculate $P(X < 0.2 \cap Y \leq 0.5)$ or $P(X < 0.2, Y \leq 0.5)$.

We know;

$$P(X < 0.2 \cap Y \leq 0.5) = \int_{0.0}^{0.2} \int_{0.0}^{0.5} f(x, y) dy dx$$

$$= \int_{0.0}^{0.2} \int_{0.0}^{0.5} 4xy dy dx$$

$$= \int_{0.0}^{0.2} 4x \left[\frac{y^2}{2} \right]_0^{0.5} dx$$

$$= \int_{0.0}^{0.2} 2x (0.5)^2 dx$$

$$= 2 \times (0.5)^2 \cdot \left[\frac{x^2}{2} \right]_0^{0.2} = (0.5)^2 \cdot (0.2)^2$$

Ques The joint pdf of two continuous R.Vs X and Y is given as:

$$f(x,y) = Axy; \quad 0 < x < 1, \quad 0 < y < x$$

(i) Find A .

(ii) Find the marginal density function of X & Y .

(iii) Check for independence of X & Y .

Ans:

(i) Total prob. is 1:

$$\int_0^1 \int_0^x Axy \, dy \, dx = 1$$

$$\Rightarrow A \cdot \int_0^1 x \left[\int_0^x y \, dy \right] dx = 1$$

$$\Rightarrow A \int_0^1 x \left[\frac{y^2}{2} \right]_0^x dx = 1$$

$$\Rightarrow A \int_0^1 x \left[\frac{x^2}{2} \right] dx = 1$$

$$\Rightarrow A \int_0^1 x^3 dx = 2$$

$$\Rightarrow A \left[\frac{x^4}{4} \right]_0^1 = 2$$

$$\Rightarrow \boxed{A = 8}$$

(ii) Finding marginal density function of X & Y .

$$f(x) = \int_0^1 f(x,y) dy$$

$$= \int_0^x 8xy dy$$

$$= 8x \left[\frac{y^2}{2} \right]_0^x$$

$$= 4x \cdot x^2$$

$$= 4x^3; \quad 0 < x < 1;$$

& $f(y) = \int_0^1 8xy dx$

$$= 8y \left[\frac{x^2}{2} \right]_0^1$$

$$= 4y; \quad 0 < y < 1$$

(if) For independence of x & y ;
 $f(x,y) = 8xy$

here limit of y
 is in terms of x .
 So, we must
 put in terms of
 y only.

& $f(x) \cdot f(y) = 4x^3 \cdot 4y = 16x^3y$

As we see $f(x,y) \neq f(x) \cdot f(y)$. Hence,
 x & y are not independent.

→ $f(y) = \int_0^1 8xy dx$;
 $= 8y \left[\frac{x^2}{2} \right]_0^1$
 $= 8y \left[\frac{1}{2} - \frac{0}{2} \right]; \quad 0 < y < 1.$

Consider this:

$$0 < x < 1$$

$$0 < y < 1$$

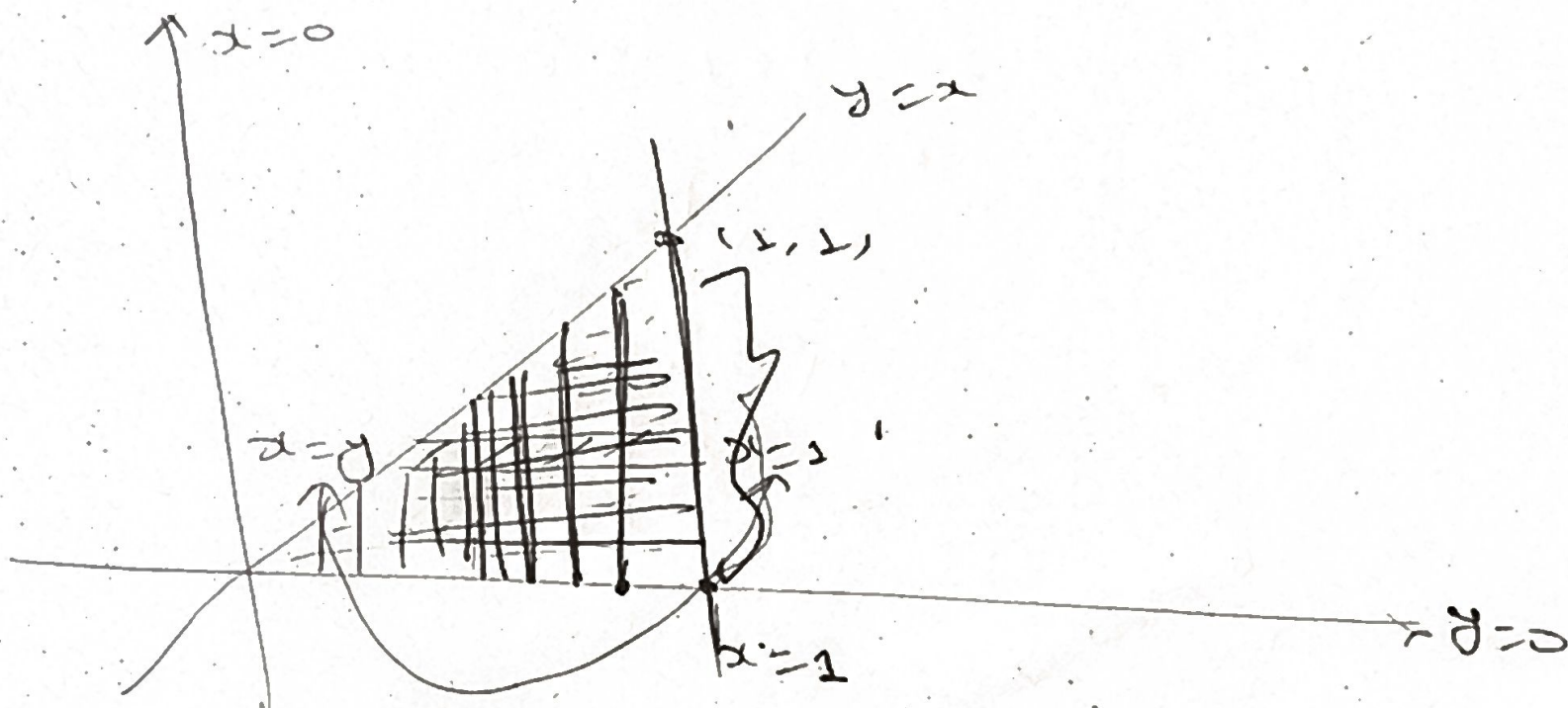


so limit
 is y to 1

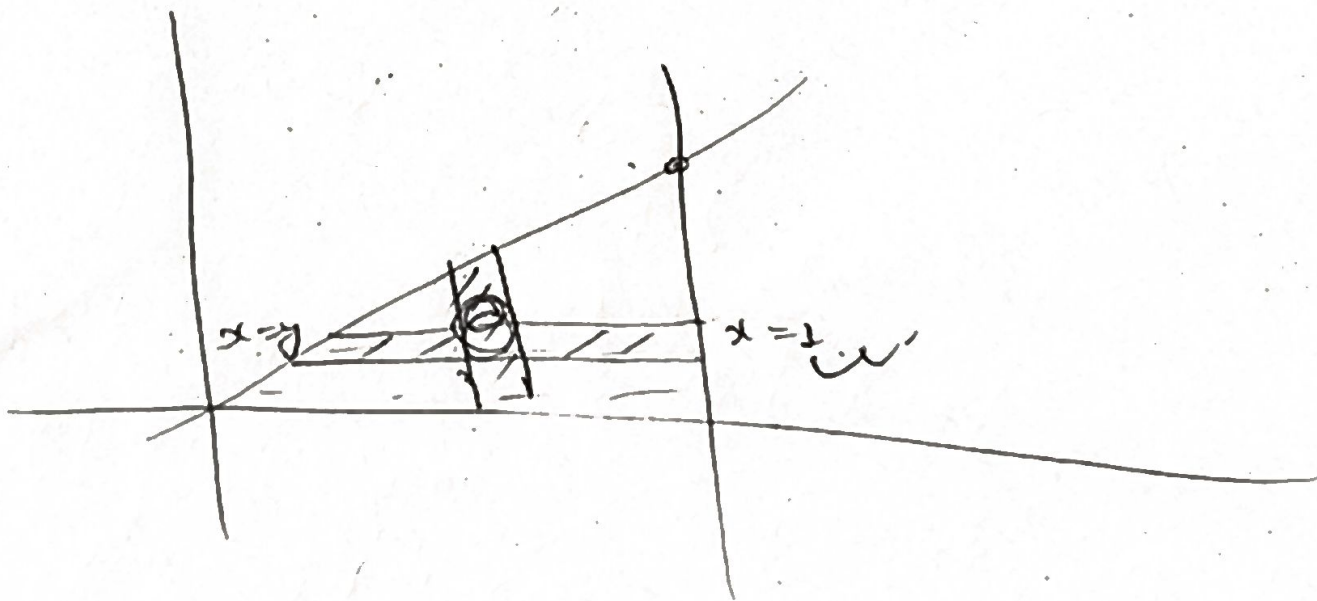
OR
 plot graph
 & see this

$$0 < x < 1; \quad x=0; \quad x=1$$

$$0 < y < 1; \quad y=0; \quad y=x$$



So limit is y to 1 for (x) .



eg; The joint pdf of two continuous r.v. x and y is given as;

$$f(x,y) = Axy; \quad 0 < x < 1, \quad 0 < y < 1$$

(i) Find A

(ii) Find the marginal density function of x & y .

(iii) Check of independence of x & y .

5.13
⑤

$$\int_0^1 \int_0^y A xy \, dx \, dy = 1$$

$$\Rightarrow \int_0^1 Ay \left[\frac{x^2}{2} \right]_0^y dy = 1$$

$$\Rightarrow A \int_0^1 \frac{y^3}{2} dy = 1$$

$$\Rightarrow A \left[\frac{y^4}{8} \right]_0^1 = 1$$

$$\therefore \boxed{A=8}$$

⑥ i. $f(x) = \int_0^1 f(x,y) dy$

$$= \int_0^1 8xy \, dy$$

$$= 8x \left[\frac{y^2}{2} \right]_0^1$$

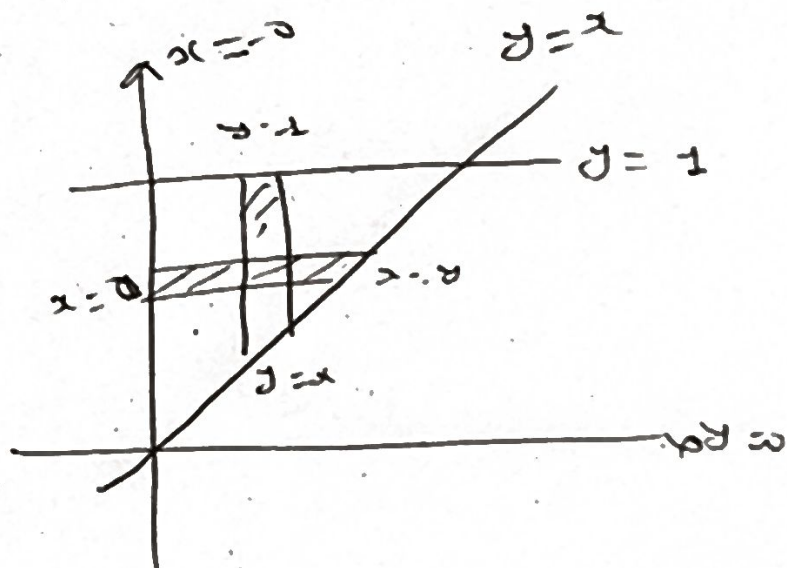
$$= 8x \left[\frac{1}{2} - \frac{x^2}{2} \right]_x; 0 < x < 1$$

$f(y) = \int_0^1 f(x,y) dx$

$$= \int_0^y 8xy \, dx$$

$$= 8y \left[\frac{x^2}{2} \right]_0^y$$

$$= 4y^3; 0 < y < 1$$



(11),

$$f(x), f(y) = (4x - 4y^2) \cdot 40^3$$

$$= 16y^3x - 16x^2y^3 \neq f(y, x)$$

So, not independent

eg, The joint pdf of two continuous r.v. X and Y is given as:

$$f(x,y) = A(6-x-y); \quad 0 < x < 2, 2 < y < 4$$

(i) Find A .

(ii) Find the marginal density function of x and y .

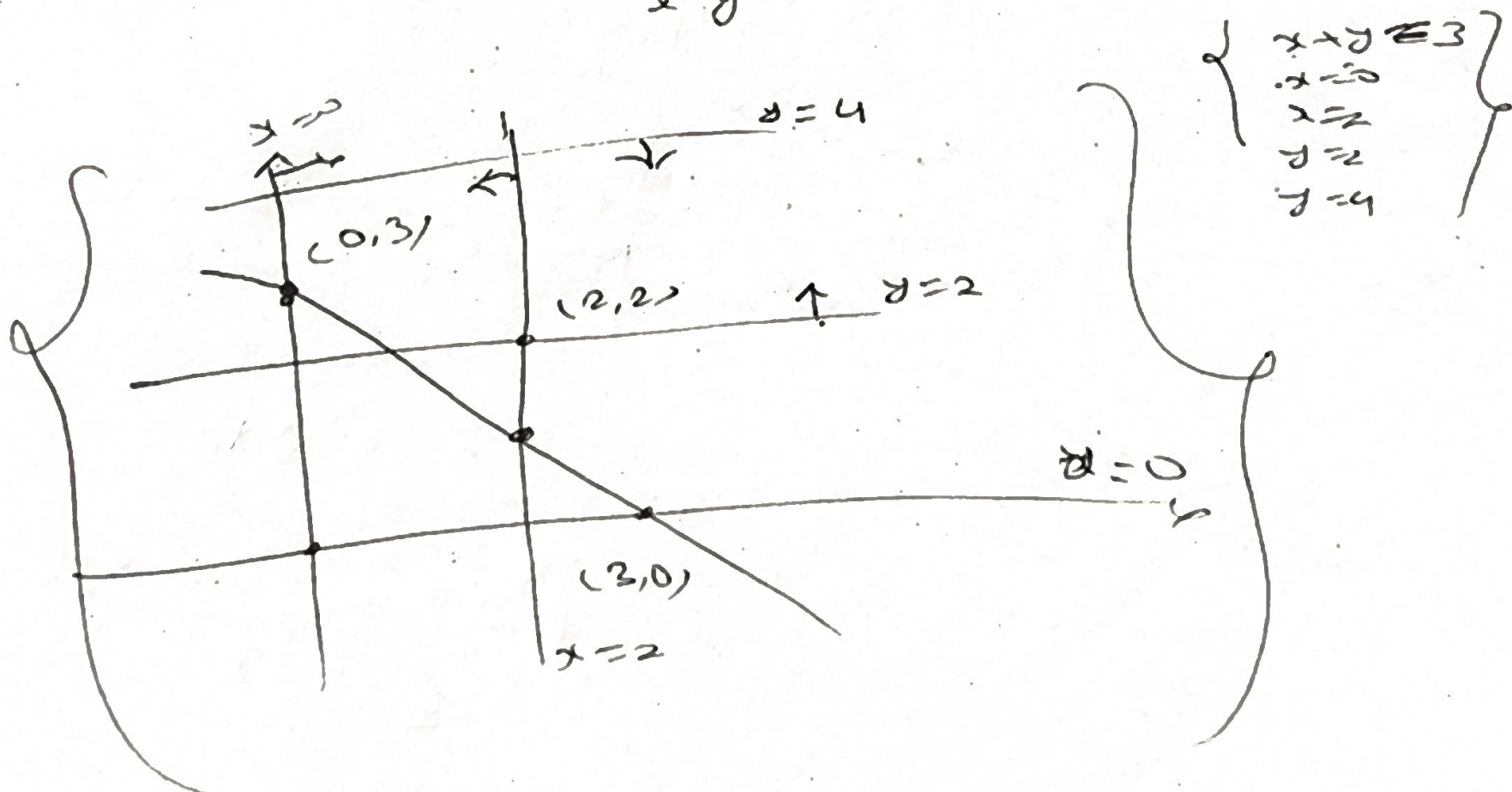
(iii) Check for independence of x & y .

(iv) Find $P(X < 1, Y < 3)$ and $P(X+Y < 3)$

(i) $A = 2/8$

(iv) $f(x,y) = \frac{1}{8}(6-x-y); \quad 0 < x < 2, 2 < y < 4$

$$P(X+Y < 3) = \int \int_{x+y < 3} f(x,y) dx dy$$



$$= \int_0^3 \left[\int_0^{3-y} \frac{1}{8} (6-x-y) dx \right] dy$$

$$= \int_0^3 \left[\frac{6}{8}x - \frac{x^2}{2} - \frac{y}{8}x \right]_0^{3-y} dy$$

$$= \int_0^3 \left[\frac{6}{8}(3-y) - \frac{(3-y)^2}{2} - \frac{y}{8}(3-y) \right] dy$$

$$= \int_0^3 \left[\frac{18}{8} - \frac{6}{8}y - \frac{9-6y+y^2}{2} - 3y + y^2 \right] dy$$

$$= \left[\frac{18}{8}y - \frac{6}{8} \cdot \frac{y^2}{2} - \frac{9y - 3y^2 - \frac{y^3}{3}}{2} - \frac{3y^2}{2} + \frac{y^3}{3} \right]_0^3$$